CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 12: Reflection

Previous year board questions, topic-wise: 7 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Reflection of figures

1. Plot $A(1, 3)$, $B(1, 2)$ and $C(3, 0)$ on graph paper. (a) Reflect A and B in the x-axis to get E and D. (b) Reflect A and B in the origin to get F and G. (c) Reflect A, B and C in the y-axis to get J, I and H. (d) Join $A, B, C, D, E, F, G, H, I, J$ in order and name the closed figure. ICSE 2025, 5 marks

2. Plot $A(0, 3)$, $B(2, 1)$ and $C(4, -1)$. (a) Reflect B and C in the y-axis to get B' and C', and write their coordinates. (b) Reflect A in the line BB' to get A', and write its coordinates. (c) Join A, B, A', B' in order and name the figure. ICSE 2024, 5 marks

3. Plot $A(0, 4)$, $B(1, 1)$ and $C(4, 0)$. (a) Reflect A and B in the x-axis to get E and D. (b) Reflect B in the origin to get F, and write its coordinates. (c) Reflect B and C in the y-axis to get H and G. (d) Join $A, B, C, D, E, F, G, H, A$ in order and name the closed figure. ICSE 2023, 5 marks

4. Plot $A(-4, 0)$, $B(-3, 2)$, $C(0, 4)$, $D(4, 1)$ and $E(7, 3)$. Reflect B, C, D and E in the x-axis to get B', C', D' and E'. Join $A, B, C, D, E, E', D', C', B', A$ in order and name the closed figure. ICSE 2020, 4 marks

5. Plot $A(0, 5)$, $B(3, 0)$, $C(1, 0)$ and $D(1, -5)$. Reflect B, C and D in the y-axis to get B', C' and D', and write their coordinates. Join $A, B, C, D, D', C', B', A$ in order and name the closed figure. ICSE 2019, 4 marks

6. Plot $A(0, 4)$, $B(2, 3)$, $C(1, 1)$ and $D(2, 0)$. Reflect B, C and D in the y-axis to get B', C' and D', and write their coordinates. Join $A, B, C, D, D', C', B', A$ in order and write the equation of the line of symmetry of the figure. ICSE 2017, 5 marks

2 Invariant points

7. $ABCD$ is a quadrilateral with vertices $A(2, 2)$, $B(2, -2)$, $C(0, -1)$ and $D(0, 1)$. (a) Reflect $ABCD$ in the y-axis to get $A'B'CD$, and write the coordinates of A' and B'. (b) Name two points that are invariant under this reflection. (c) Name the polygon $A'B'CD$. ICSE 2018, 4 marks

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